Failure Detection and Heartbeats

how a cluster notices a dead node

Phase 4 videos on YouTube

In one line: nodes send periodic heartbeats and a node that stays silent past a timeout is suspected dead; there's no perfect detector, so you trade detection speed against false alarms, often with adaptive detectors like phi accrual and a suspicion phase before acting.

1 The question every cluster must answer

In a cluster of fifty nodes, one dies at 3 a.m. How fast do the others notice? Too slow, and requests keep going to a dead node. Too fast, and a healthy node that just paused for garbage collection gets thrown out, triggering a failover nobody needed. A **failure detector** is the component that makes this call.

2 Heartbeats and timeouts

The simplest detector: each node sends a small **heartbeat** message at a fixed interval (say every second). If a peer hears nothing for longer than a **timeout** (say 3 seconds), it marks that node as suspected dead. The timeout is a guess: the node may be dead, slow, paused, or cut off by the network (lesson 3).

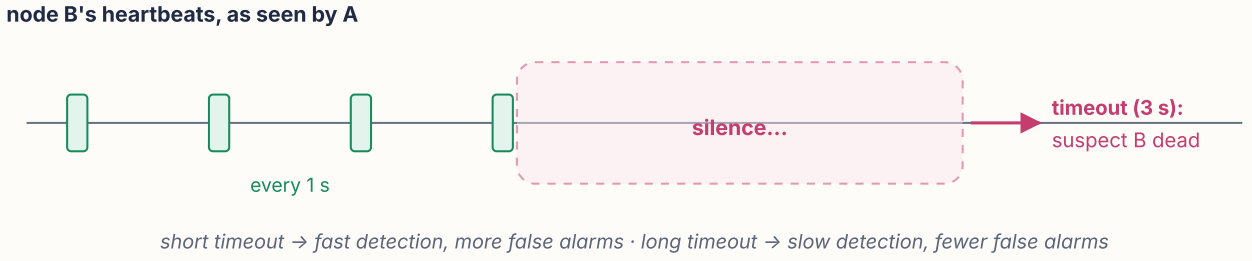


Figure 1. After a few missed heartbeats, the silent node is suspected. The timeout length is the key trade-off.

3 No perfect detector

A detector is judged on two properties: **completeness** (every crashed node is eventually suspected) and **accuracy** (healthy nodes aren't wrongly suspected). In a real network with unbounded delays you can't have both perfectly; you can only tune the balance. That's why actions after detection must be safe even when the detector is wrong.

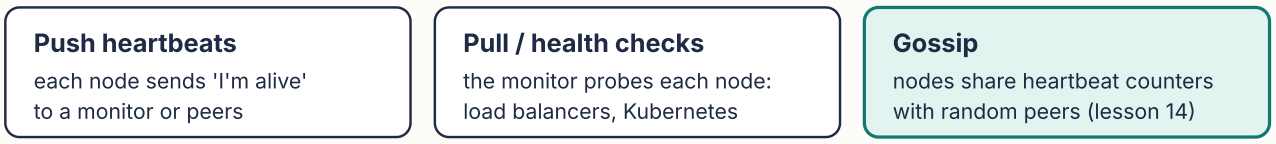


Figure 2. Three ways to collect liveness information. Large clusters prefer gossip because no single monitor sees everything.

4 Adaptive detection: phi accrual

A fixed timeout ignores how the network behaves. The **phi accrual** detector keeps a window of recent heartbeat intervals, and computes a suspicion level **φ** (phi) that grows the longer the silence lasts compared with what's normal for that node. Above a threshold (Cassandra's default is 8), the node is marked down. On a jittery link, φ grows slowly; on a steady one, a short silence is already suspicious.

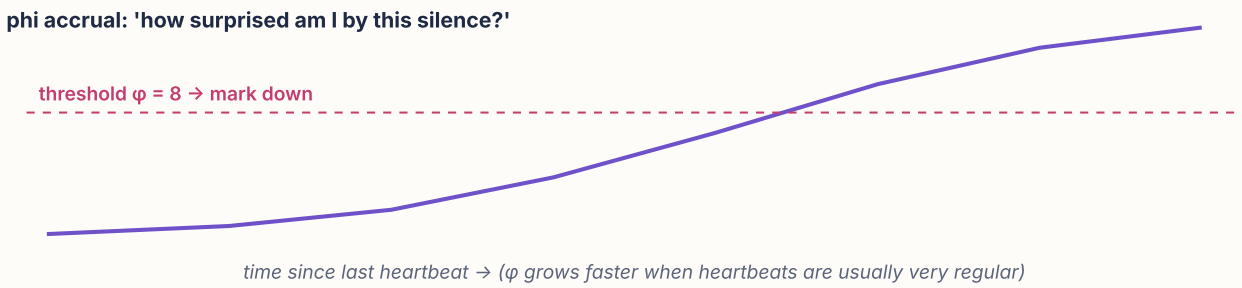


Figure 3. φ rises with the length of the silence; crossing the threshold marks the node down.

WORKED EXAMPLE: CASSANDRA

Cassandra nodes gossip every second, exchanging heartbeat counters for every node they know about. Each node keeps a window of heartbeat arrival intervals for every peer. When node 7's heartbeats stop, φ for node 7 climbs on each observer; past 8, they mark it down locally, stop sending it reads, and store **hints** for writes it misses. If node 7 was only paused by a long GC, its heartbeats return, φ falls, and it's marked up again, with no failover drama, because Cassandra has no single leader to replace.

5 Making detection safe

- Use a **suspicion phase** (SWIM, Consul): ask other nodes to probe the suspect before declaring it dead.
- Make the cost of a false positive small: a node marked dead should be able to rejoin cleanly.
- For leaders, combine detection with **leases and fencing** (lessons 8 and 13) so a wrongly-suspected leader can't do harm.
- Separate **liveness** (is it running?) from **readiness** (should it get traffic?).

COMMON MISTAKES

- Timeouts shorter than a normal GC pause or deploy restart.
- Heartbeats on the same congested path as data, so load looks like death.
- Failing over immediately on one missed heartbeat.
- A single central monitor that itself can fail or be partitioned.

6 Key terms

Heartbeat: periodic 'I'm alive' message

Failure detector: decides which nodes are suspected dead

Completeness: crashed nodes are eventually suspected

Accuracy: healthy nodes are not suspected

Phi accrual: adaptive detector giving a suspicion level, not yes/no

SWIM: gossip-based membership with indirect probes

7 Interview questions

Q1. How do heartbeats detect failure? BASIC

Each node sends periodic heartbeats; if one isn't heard for longer than a timeout, it is suspected dead. The timeout trades detection speed for false alarms.

Q2. Why can't failure detection be perfect? BASIC

Network delays have no fixed upper bound, so a slow or partitioned node looks exactly like a dead one. You can only balance speed against wrong suspicions.

Q3. A healthy node keeps getting marked dead during GC pauses. What do you change? SCENARIO

Raise the timeout above the worst normal pause, use an adaptive detector like phi accrual, add indirect probes before declaring death, and tune GC to shorten pauses.

Q4. How fast should a database cluster fail over? SCENARIO

Fast enough for the availability target, slow enough to ride out normal blips: typically a few to tens of seconds, plus fencing so a wrongly-replaced leader can't keep writing.

Q5. What is the risk of a too-aggressive failure detector? TRICKY

False positives trigger needless failovers and rebalancing, which add load and can cause more false positives: a flapping cluster. In leader systems it can create two leaders without fencing.

Q6. Why do large clusters use gossip for failure detection? TRICKY

There is no single monitor to overload or partition; every node learns about failures from random peers in O(log n) rounds, and indirect probing reduces false alarms.

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